

## PAKRI, Estonia

WMO: 260290

Lat: 59.3894N

Lon: 24.0400E

Elev: 80

StdP: 14.65

Time Zone: 2.00 (EUE)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 3.5        | 9.0 | -3.1                          | 4.7 | 4.9  | 2.5 | 6.3 | 9.7  | 29.6                  | 31.8 | 27.3 | 31.3 | 7.3                   | 120  | 0.637 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.6                    | 76.9            | 66.9 | 73.4 | 65.8 | 70.5 | 63.9 | 68.6                | 74.9 | 66.6 | 72.0 | 64.8 | 69.5 | 5.6                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 66.1                            | 96.5 | 72.2 | 64.4 | 90.9 | 70.2 | 62.7 | 85.6 | 68.5 | 32.9          | 75.1 | 31.3 | 72.3 | 29.9 | 69.5 |                   |
|                                 |      |      |      |      |      |      |      |      |               |      |      |      |      |      |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 28.3          | 24.8 | 21.8 | 0.0                        | 83.2 | 9.3                | 3.8 | -6.7                                               | 86.0 | -12.2      | 88.2 | -17.4      | 90.4 | -24.1      | 93.2 |    |    |
| (5)               |      |      | -0.6                       | 71.3 | 8.7                | 2.0 | -6.9                                               | 72.7 | -12.0      | 73.9 | -16.9      | 75.0 | -23.2      | 76.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 44.2        | 26.8  | 26.5  | 31.3  | 39.4  | 49.1  | 56.9  | 63.9  | 63.0  | 55.8  | 45.5  | 37.9  | 32.6  |  |  |
|                                                                        | DBStd       | 14.54       | 9.30  | 8.60  | 6.24  | 5.31  | 6.14  | 4.88  | 4.73  | 4.40  | 4.58  | 6.35  | 6.21  | 8.33  |  |  |
|                                                                        | HDD50       | 3444        | 718   | 659   | 579   | 321   | 89    | 3     | 0     | 0     | 9     | 163   | 362   | 541   |  |  |
|                                                                        | HDD65       | 7687        | 1183  | 1079  | 1044  | 768   | 494   | 249   | 77    | 92    | 278   | 605   | 812   | 1006  |  |  |
|                                                                        | CDD50       | 1314        | 0     | 0     | 0     | 4     | 61    | 210   | 432   | 402   | 181   | 24    | 0     | 0     |  |  |
|                                                                        | CDD65       | 81          | 0     | 0     | 0     | 0     | 2     | 6     | 43    | 29    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 267         | 0     | 0     | 0     | 0     | 12    | 21    | 138   | 94    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 39          | 0     | 0     | 0     | 0     | 1     | 4     | 19    | 15    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 9.8         | 10.7  | 10.3  | 10.5  | 9.0   | 8.4   | 9.3   | 8.3   | 8.3   | 9.6   | 10.7  | 10.7  | 12.1  |  |  |
|                                                                        | PrecAvg     | 25.0        | 2.0   | 1.6   | 1.2   | 1.3   | 1.3   | 2.2   | 2.7   | 2.9   | 2.3   | 2.8   | 2.6   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecMax     | 39.8        | 3.5   | 3.5   | 2.3   | 3.1   | 3.3   | 4.7   | 7.6   | 6.1   | 5.5   | 5.2   | 4.8   | 4.1   |  |  |
|                                                                        | PrecMin     | 15.3        | 0.2   | 0.2   | 0.2   | 0.1   | 0.3   | 0.3   | 0.1   | 0.4   | 0.7   | 0.6   | 0.2   | 0.6   |  |  |
|                                                                        | PrecStd     | 4.8         | 0.8   | 0.9   | 0.6   | 0.8   | 0.7   | 1.1   | 1.5   | 1.4   | 1.3   | 1.3   | 1.1   | 0.9   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 42.2  | 41.4  | 49.4  | 60.6  | 74.6  | 76.6  | 82.7  | 82.5  | 70.7  | 59.3  | 50.7  | 47.4  |  |  |
|                                                                        |             | MCWB        | 40.6  | 39.2  | 41.2  | 47.6  | 61.9  | 64.4  | 70.3  | 68.4  | 61.6  | 55.1  | 48.9  | 45.5  |  |  |
|                                                                        | 2%          | DB          | 40.3  | 38.9  | 43.8  | 54.4  | 65.8  | 70.2  | 77.6  | 76.2  | 66.4  | 57.1  | 48.7  | 44.9  |  |  |
|                                                                        |             | MCWB        | 39.0  | 37.7  | 38.6  | 45.7  | 56.4  | 62.5  | 68.3  | 66.7  | 60.3  | 54.1  | 47.1  | 42.9  |  |  |
|                                                                        | 5%          | DB          | 38.7  | 37.3  | 40.1  | 50.7  | 61.3  | 66.7  | 74.3  | 72.6  | 63.9  | 55.5  | 47.3  | 43.3  |  |  |
|                                                                        |             | MCWB        | 37.3  | 36.0  | 37.2  | 43.9  | 53.4  | 60.1  | 67.3  | 65.1  | 59.2  | 52.9  | 45.6  | 41.5  |  |  |
|                                                                        | 10%         | DB          | 37.3  | 35.9  | 38.1  | 47.5  | 57.9  | 64.0  | 71.4  | 69.8  | 62.1  | 53.8  | 45.9  | 41.6  |  |  |
|                                                                        |             | MCWB        | 36.0  | 34.9  | 35.9  | 42.1  | 51.6  | 58.4  | 65.3  | 63.7  | 57.9  | 51.4  | 44.1  | 39.9  |  |  |
|                                                                        | 0.4%        | WB          | 41.3  | 39.9  | 42.5  | 50.2  | 62.2  | 66.5  | 71.8  | 70.1  | 63.3  | 56.2  | 49.3  | 45.8  |  |  |
|                                                                        |             | MCDB        | 42.0  | 40.9  | 48.6  | 56.4  | 72.9  | 71.7  | 79.4  | 76.9  | 67.7  | 58.0  | 50.2  | 46.8  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 39.4  | 37.8  | 39.4  | 46.7  | 57.7  | 63.9  | 69.7  | 67.9  | 61.4  | 54.8  | 47.6  | 43.4  |  |  |
|                                                                        |             | MCDB        | 40.2  | 38.5  | 42.4  | 52.6  | 64.0  | 69.6  | 76.1  | 74.3  | 65.4  | 56.6  | 48.4  | 44.6  |  |  |
|                                                                        | 5%          | WB          | 37.6  | 36.3  | 37.7  | 44.6  | 54.8  | 61.0  | 67.6  | 66.1  | 59.8  | 53.3  | 45.9  | 41.7  |  |  |
|                                                                        |             | MCDB        | 38.5  | 37.0  | 39.8  | 50.0  | 60.0  | 65.4  | 73.3  | 71.1  | 63.4  | 55.1  | 46.9  | 43.0  |  |  |
|                                                                        | 10%         | WB          | 36.1  | 35.0  | 36.2  | 42.6  | 52.2  | 58.8  | 65.9  | 64.5  | 58.4  | 51.8  | 44.4  | 40.1  |  |  |
|                                                                        |             | MCDB        | 37.1  | 35.8  | 37.8  | 46.8  | 56.6  | 63.1  | 70.9  | 68.9  | 61.6  | 53.7  | 45.6  | 41.4  |  |  |
|                                                                        | 5% DB       | MDBR        | 6.3   | 6.4   | 6.9   | 8.6   | 9.6   | 8.5   | 8.6   | 9.3   | 7.6   | 6.7   | 5.4   | 5.8   |  |  |
|                                                                        |             | MCDBR       | 5.5   | 5.5   | 9.6   | 15.2  | 16.8  | 14.0  | 13.7  | 14.1  | 10.7  | 7.7   | 5.8   | 5.2   |  |  |
|                                                                        |             | MCWBR       | 5.6   | 4.8   | 6.9   | 8.9   | 9.6   | 8.2   | 8.0   | 7.9   | 7.0   | 5.8   | 5.8   | 5.3   |  |  |
|                                                                        |             | MCWBR       | 5.6   | 5.2   | 8.6   | 13.4  | 14.8  | 12.3  | 12.3  | 11.7  | 9.5   | 6.5   | 5.5   | 5.1   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR       | 5.7   | 4.9   | 6.5   | 8.5   | 9.6   | 8.4   | 7.7   | 7.6   | 7.0   | 5.6   | 5.7   | 5.4   |  |  |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|                                                                        | taub        |             | 0.280 | 0.311 | 0.333 | 0.370 | 0.365 | 0.363 | 0.390 | 0.384 | 0.353 | 0.335 | 0.313 | 0.269 |  |  |
|                                                                        |             | taud        | 2.157 | 2.205 | 2.264 | 2.311 | 2.403 | 2.458 | 2.407 | 2.419 | 2.489 | 2.408 | 2.226 | 2.108 |  |  |
| (42) Clear-Sky Solar Irradiance                                        | Ebn at Noon |             | 160   | 218   | 253   | 263   | 274   | 277   | 265   | 257   | 248   | 212   | 146   | 118   |  |  |
|                                                                        |             | Edh at Noon | 15    | 23    | 30    | 34    | 33    | 32    | 33    | 30    | 24    | 19    | 14    | 10    |  |  |
|                                                                        | RadAvg      |             | 81    | 253   | 735   | 1251  | 1761  | 1848  | 1754  | 1381  | 842   | 365   | 100   | 51    |  |  |
| (45) All-Sky Solar Radiation                                           | RadStd      |             | 13    | 42    | 89    | 135   | 180   | 120   | 146   | 133   | 103   | 48    | 17    | 7     |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (54 neighbors) |  | +0.72 | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | -243        | N/A   | N/A   | N/A   |

Nomenclature: See separate page